

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	01 September 2026
Team ID	Individual Project
Project Name	Software Installation Request Automation in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architectural diagram and the information in Table-1 and Table-2, adapted to the ServiceNow software installation request automation project.

Example: Software installation request processing through ServiceNow Service Catalog and Workflow/Automation

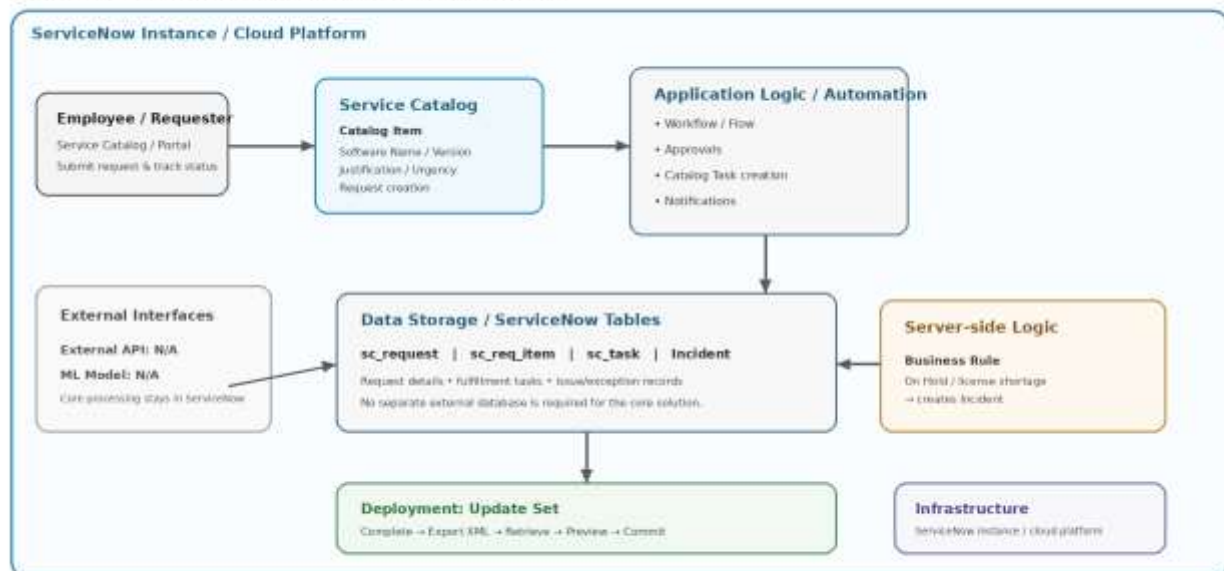
Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

Guidelines:

- Include all the processes as application logic / technology blocks.
- Provide infrastructural demarcation (ServiceNow instance / cloud platform).
- Indicate external interfaces (External API: N/A for the core solution).
- Indicate data storage components / services.
- Indicate interface to machine learning models: N/A because no ML model is used.

Technical Architecture - Software Installation Request Automation

ServiceNow application logic, data storage, integrations and deployment boundary



Flow: Requester -> Service Catalog -> Automation/Approvals -> Fulfillment & Data Tracking -> Exception Handling -> Deployment

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How the user interacts with the application: Service Catalog and Service Portal request experience.	ServiceNow Service Catalog / Service Portal
2	Application Logic-1	Logic for software request intake, validation and request creation.	ServiceNow Catalog Item + Catalog Variables
3	Application Logic-2	Logic for approval and controlled request processing.	ServiceNow Workflow / Flow Automation
4	Application Logic-3	Logic for IT fulfillment task creation, assignment and notifications.	ServiceNow Catalog Tasks + Notifications
5	Database	Data types and configuration records for requests and fulfillment.	ServiceNow tables: sc_request, sc_req_item, sc_task
6	Cloud Database	Cloud database service used by the application.	ServiceNow platform data store (managed by the platform)
7	File Storage	File storage requirements.	N/A for core project functionality
8	External API-1	Purpose of external API used in the application.	N/A - no external API in core solution
9	External API-2	Purpose of external API used in the application.	N/A - no external API in core solution
10	Machine Learning Model	Purpose of machine learning model.	N/A - no ML model is used
11	Infrastructure (Server / Cloud)	Application deployment environment.	ServiceNow instance / cloud platform

Local Server Configuration: N/A - the core application is configured in the ServiceNow instance.

Cloud Server Configuration: ServiceNow instance / cloud platform.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No external open-source application framework is required for the core solution.	N/A - ServiceNow platform
2	Security	ServiceNow authentication, roles/groups and ACL-based access control can restrict records and configuration. Approval routing provides controlled authorization.	ServiceNow Authentication, Roles/Groups, ACLs, approval controls
3	Scalable Architecture	The solution uses platform-managed application services and table-based processing. Catalog requests, tasks and automation are separated logically by function.	ServiceNow platform, Service Catalog, Workflow/Flow
4	Availability	The application runs within the ServiceNow instance and uses platform-managed application services rather than a self-hosted local server.	ServiceNow platform
5	Performance	The design uses structured catalog variables, automated task creation and table-based tracking to reduce manual processing. Repeated request testing validates consistent execution.	ServiceNow Service Catalog, Workflow/Flow, server-side Business Rule

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>